The variance of squarefree polynomial values

Omer Simhi

For a polynomial $P \in \mathbb{Z}[t]$ we define

$$E_P(x) := \sum_{n \le x} \mu^2(P(n)) - \mathfrak{S}_P x, \qquad \mathfrak{S}_P := \prod_p \left(1 - \frac{\rho_P\left(p^2\right)}{p^2}\right),$$

where $\rho_P(q) := \#\{n \bmod q : P(n) \equiv 0 \bmod q\}$. Sofos recently proved the following. Below, d and x are positive integers.

Theorem 1 (Sofos [Sof26])**.** *Fix $d \ge 3$. Let*

$$\mathscr{P}_d(H) := \{P \in \mathbb{Z}[t] : \deg P = d,\ |P| \le H\}, \qquad V_d(x,H) := \frac{1}{(2H)^{d+1}} \sum_{P \in \mathscr{P}_d(H)} E_P(x)^2.$$

Here $|P| := \max_{0 \le j \le d} |a_j(P)|$ ($a_j(P)$ being the coefficients of P). Let

$$\gamma_0 := -4\frac{\zeta\left(-\frac{1}{2}\right)}{\zeta(2)^2} \prod_p \left(1 - \frac{2 - p^{-\frac{1}{2}} + p^{-1}}{(p+1)^2}\right) = 0.192\ldots. \tag{1}$$

Fix $\varepsilon > 0$. For $x \le H^{1/(d+3)-\varepsilon}$ we have

$$V_d(x,H) = \gamma_0 x^{\frac{1}{2}} + O_\varepsilon\left(x^{\frac{1}{2}-\delta}\right)$$

for $\delta = \delta_{d,\varepsilon} > 0$. For $0 < \varepsilon < 1/18$ and $x \le H^{1/(d+28/9)-\varepsilon}$, one can take $\delta = 1/18 - \varepsilon$.

Sofos' argument uses complex analysis and correlation of squarefree values, and is very much in the spirit of an earlier argument of R. R. Hall [Hal82] on squarefree numbers in short intervals. The purpose of this note is to give a simpler, elementary proof of the variance asymptotic in Theorem 1, avoiding complex analysis and correlation estimates. Our approach extends the result to $d = 1, 2$ and gives the sharper error $O(x^{1/3})$, albeit in a more restrictive range of H. Sofos notes [Sof26, Remark 5.1] that even the Lindelöf hypothesis does not improve the error exponent from his contour-integration argument.

Our starting point is a representation for the second moment of E_P considered by Theorem 1; this representation is derived in our paper on the distribution of squarefree values of random polynomials [Sim26]. We need to introduce notation to explain it.

For $q \ge 1$, let

$$V_q := \{g \in (\mathbb{Z}/q\mathbb{Z})[t] : \deg g \le d\}.$$

For $k_1, k_2, \ell_1, \ell_2 \ge 1$, put $q := [k_1^2, k_2^2]$, and define

$$\mathfrak{A}_{\mathbf{k}}(\boldsymbol{\ell}) := \frac{1}{q^{d+1}} \sum_{\nu_i \bmod k_i^2\ (i=1,2)} e\left(-\frac{\ell_1\nu_1}{k_1^2} - \frac{\ell_2\nu_2}{k_2^2}\right) \#\left\{g \in V_q : g(\nu_i) \equiv 0 \bmod k_i^2,\ i = 1,2\right\}, \tag{2}$$

where $e(t) := e^{2\pi i t}$. Moreover, let

$$E_x(\theta) := \sum_{n \le x} e(n\theta).$$

Put

$$h_0(d) := \max\left\{\frac{11}{2}, d+3, \frac{6d+11}{2d+1}\right\}, \qquad h_1(d) := \max\left\{\frac{19}{3}, d+\frac{10}{3}, \frac{20d+38}{6d+3}\right\}.$$

While these seem complicated, for $d \ge 3$ we have $h_0(d) = d+3$ and $h_1(d) = d + 10/3$.

Proposition 2. *[Sim26] Fix $d \geq 1$ and $\varepsilon > 0$. Uniformly for $H \geq x \geq 2$ and $1 < z \leq \sqrt{2H}$,*

$$V_d(x,H) = M(x) + O_{d,\varepsilon}\left(\frac{x^{2+\varepsilon}H^{\varepsilon}}{z^{1-\varepsilon}} + x^{2+\varepsilon}H^{-1/3+\varepsilon} + x^{2+d/2+\varepsilon}H^{-1/2+\varepsilon} + x^2 z^2\left(\frac{\log x}{H} + \frac{x}{H^2}\right) + \frac{z^{4+\varepsilon}}{H} + \frac{z^{2d+4+\varepsilon}}{H^{d+1/2}}\right) \quad (3)$$

where

$$M(x) := \sum_{k_1,k_2\geq 1} \frac{\mu(k_1)\mu(k_2)}{k_1^2k_2^2} \sum_{\substack{0<\ell_i<k_i^2\ (i=1,2)\\ \ell_1/k_1^2+\ell_2/k_2^2\in\mathbb{Z}}} E_x\left(\frac{\ell_1}{k_1^2}\right) E_x\left(\frac{\ell_2}{k_2^2}\right) \mathfrak{A}_{\mathbf{k}}(\boldsymbol{\ell}).$$

In particular, for each fixed $\eta > 0$,

$$V_d(x,H) = M(x) + O_{d,\eta}(x^{1/2-\delta}) \quad \text{if} \quad H \geq x^{h_0(d)+\eta},$$
$$V_d(x,H) = M(x) + O_{d,\eta}(x^{1/3}) \quad \text{if} \quad H \geq x^{h_1(d)+\eta}$$

Proof. This result is implicit in [Sim26]; we fill in the details. First, identify $M(x)$ with the local second moment from [Sim26, Prop. 4.1]. Using the finite-prime Möbius expansion [Sim26, Eq. (4.3)] and orthogonality over the full residue spaces V_q gives the coefficients $\mathfrak{A}_{\mathbf{k}}(\boldsymbol{\ell})$ in (2). The convergence arguments of [Sim26, Lems. 4.2–4.3] justify passing to the infinite sum. Thus, with the notation of [Sim26, Section 4],

$$\frac{1}{\zeta(2d+2)} \int_{V_{\square}^{\sharp}} E_g(x)^2\, dg = M(x).$$

Applying [Sim26, Lems. 2.4] in [Sim26, Prop. 2.3], followed by [Sim26, Prop. 4.1], gives (3). For the last assertions, choose $z = x^{3/2+\kappa}(H/x^{h_0(d)+\eta})^{1/10}$ (resp. $z = x^{5/3+\kappa}(H/x^{h_1(d)+\eta})^{1/10}$) where $\kappa > 0$ is sufficiently small in terms of d and η. The definition of $h_0(d)$ (resp. $h_1(d)$) ensures that every error term in (3) is $O_{d,\eta}(x^{1/2-\delta})$ (resp. $O_{d,\eta}(x^{1/3})$) provided $H \geq x^{h_0(d)+\eta}$ (resp. $H \geq x^{h_1(d)+\eta}$) and ε is sufficiently small. □

Our main result is the following.

Theorem 3. *Fix $d \geq 1$. Then*

$$M(x) = \widetilde{\gamma_0} x^{1/2} + O_d(x^{1/3}),$$

where

$$\widetilde{\gamma_0} := \frac{\zeta(3/2)}{\pi\zeta(2)^2} \prod_p \left(1 - \frac{1}{p}\right)\left(1 + \frac{B(p)}{p} + \frac{B(p)}{p^{5/2}}\right), \qquad B(p) := \frac{p^3}{(p-1)(p+1)^2} = 1 + O(p^{-1}).$$

The functional equation of ζ and elementary simplification of the Euler factors give $\widetilde{\gamma_0} = \gamma_0$. From Proposition 2 and Theorem 3 we obtain at once the following corollary.

Corollary 4. *For every fixed $\eta > 0$, there is $\delta = \delta_{d,\eta} > 0$ such that*

$$V_d(x,H) = \gamma_0 x^{1/2} + O_{d,\eta}(x^{1/2-\delta}) \qquad \text{if } H \geq x^{h_0(d)+\eta}, \quad (4)$$
$$V_d(x,H) = \gamma_0 x^{1/2} + O_{d,\eta}(x^{1/3}) \qquad \text{if } H \geq x^{h_1(d)+\eta}. \quad (5)$$

Proof of Theorem 3

We begin by evaluating the inner polynomial count in (2).

Lemma 5. *Fix $d \geq 1$. For $k_1, k_2 \geq 1$ and $\nu_i \in \mathbb{Z}/k_i^2\mathbb{Z}$,*

$$\#\left\{g \in V_q : g(\nu_i) \equiv 0 \bmod k_i^2,\ i = 1,2\right\} = \frac{q^{d+1}\gcd(k_1^2, k_2^2, \nu_1 - \nu_2)}{k_1^2k_2^2}.$$

Proof. Let

$$\Phi\colon V_q \longrightarrow \mathbb{Z}/k_1^2\mathbb{Z} \times \mathbb{Z}/k_2^2\mathbb{Z}, \qquad \Phi(g) := (g(\nu_1), g(\nu_2)).$$

Then

$$\#\left\{g \in V_q : g(\nu_i) \equiv 0 \bmod k_i^2,\ i = 1,2\right\} = |\ker\Phi| = \frac{q^{d+1}}{|\operatorname{Im}\Phi|}.$$

We determine the image prime by prime. For $p \mid q$, write

$$\alpha := v_p(k_1^2), \qquad \beta := v_p(k_2^2), \qquad e := \max(\alpha, \beta),$$

and define

$$\Phi_p \colon V_{p^e} \longrightarrow \mathbb{Z}/p^\alpha\mathbb{Z} \times \mathbb{Z}/p^\beta\mathbb{Z}, \qquad \Phi_p(g) := (g(\nu_1), g(\nu_2)).$$

By the Chinese remainder theorem,

$$|\operatorname{Im}\Phi| = \prod_{p|q} |\operatorname{Im}\Phi_p|.$$

Assume without loss of generality that $\alpha \geq \beta$, so $e = \alpha$. Writing $g(t) = \sum_{j=0}^{d} c_j (t-\nu_1)^j$ is an invertible change of coordinates modulo p^α. Thus $g(\nu_1) = c_0$, while

$$g(\nu_2) \equiv c_0 + (\nu_2 - \nu_1)X \bmod p^\beta,$$

where $X := c_1 + c_2(\nu_2 - \nu_1) + \cdots + c_d(\nu_2 - \nu_1)^{d-1}$. Since $d \geq 1$, varying c_1 shows that X ranges over all of $\mathbb{Z}/p^\beta\mathbb{Z}$. Hence, for each of the p^α possible values of $g(\nu_1)$, the number of possible values of $g(\nu_2)$ is

$$\frac{p^\beta}{\gcd(p^\beta, \nu_1 - \nu_2)}.$$

Therefore

$$|\operatorname{Im}\Phi_p| = \frac{p^{\alpha+\beta}}{\gcd(p^\alpha, p^\beta, \nu_1 - \nu_2)}.$$

Multiplying over $p \mid q$ yields

$$|\operatorname{Im}\Phi| = \frac{k_1^2 k_2^2}{\gcd(k_1^2, k_2^2, \nu_1 - \nu_2)},$$

and the result follows. □

Corollary 6. *Fix $d \geq 1$. We have*

$$M(x) = \widetilde{M}(x)$$

where

$$\widetilde{M}(x) := \sum_{k_1,k_2 \geq 1} \frac{\mu(k_1)\mu(k_2)}{k_1^2 k_2^2} \sum_{\substack{s \mid k_1^2 \\ s \mid k_2^2}} \frac{\varphi(s)}{s} \sum_{\ell=1}^{s-1} \left| E_x\left(\frac{\ell}{s}\right)\right|^2. \tag{6}$$

Proof. Substituting Lemma 5 into (2), which is then substituted into $M(x)$, we obtain

$$M(x) = \sum_{k_1,k_2 \geq 1} \frac{\mu(k_1)\mu(k_2)}{k_1^4 k_2^4} \sum_{\substack{0 < \ell_i < k_i^2 \ (i=1,2) \\ \ell_1/k_1^2 + \ell_2/k_2^2 \in \mathbb{Z}}} E_x\left(\frac{\ell_1}{k_1^2}\right) E_x\left(\frac{\ell_2}{k_2^2}\right) \sum_{\nu_i \bmod k_i^2 \ (i=1,2)} e\left(-\frac{\ell_1\nu_1}{k_1^2} - \frac{\ell_2\nu_2}{k_2^2}\right) \gcd(k_1^2, k_2^2, \nu_1 - \nu_2). \tag{7}$$

Put $r := \gcd(k_1^2, k_2^2)$. Since $0 < \ell_i < k_i^2$, the condition $\ell_1/k_1^2 + \ell_2/k_2^2 \in \mathbb{Z}$ in (7) is equivalent to $\ell_1/k_1^2 + \ell_2/k_2^2 = 1$. It follows that there is a unique $1 \leq \ell \leq r-1$ such that

$$\ell_1 = \ell\frac{k_1^2}{r}, \qquad \ell_2 = (r-\ell)\frac{k_2^2}{r}.$$

Hence

$$E_x\left(\frac{\ell_1}{k_1^2}\right) E_x\left(\frac{\ell_2}{k_2^2}\right) = \left| E_x\left(\frac{\ell}{r}\right)\right|^2.$$

Moreover, using $\ell_1/k_1^2 = \ell/r$, $\ell_2/k_2^2 = 1 - \ell/r$, the exponential factor $e\big(-\frac{\ell_1\nu_1}{k_1^2} - \frac{\ell_2\nu_2}{k_2^2}\big)$ simplifies as $e\big(\frac{\ell(\nu_2-\nu_1)}{r}\big)$. Thus both the exponential and the greatest common divisor depend only on $\nu_1, \nu_2 \bmod r$. Grouping the terms according to $h = \nu_2 - \nu_1 \bmod r$, we obtain that the inner sum in (7) is

$$\begin{aligned}\sum_{\nu_i \bmod k_i^2 \ (i=1,2)} e\left(-\frac{\ell_1\nu_1}{k_1^2} - \frac{\ell_2\nu_2}{k_2^2}\right) \gcd(k_1^2, k_2^2, \nu_1 - \nu_2) &= \frac{k_1^2 k_2^2}{r} \sum_{h \bmod r} e\left(\frac{h\ell}{r}\right) \gcd(r, h) \\ &= \frac{k_1^2 k_2^2}{r} \sum_{c \mid r} \varphi(c) \sum_{u \bmod r/c} e\left(\frac{u\ell}{r/c}\right) = k_1^2 k_2^2 \sum_{\substack{v \mid r \\ v \mid \ell}} \frac{v}{r}\, \varphi\left(\frac{r}{v}\right),\end{aligned}$$

where in the last two lines we used $\gcd(r,h)=\sum_{c\mid r,\,c\mid h}\varphi(c)$, orthogonality of additive characters, and then wrote $v=r/c$. Substituting these evaluations into (7), summing first over ℓ, and writing $\ell=vL$, we obtain $M(x)=\widetilde{M}(x)$ after relabeling variables. □

The ℓ-sum in (6) is a standard calculation and is left to the reader.

Lemma 7. *For every integer $q\ge 1$,*

$$\sum_{\ell=1}^{q-1}\left|E_x\left(\frac{\ell}{q}\right)\right|^2 = q^2\left\{\frac{x}{q}\right\}\left(1-\left\{\frac{x}{q}\right\}\right).$$

Set

$$F(t):=\{t\}(1-\{t\}).$$

Then $F(t)\ll\min(1,t)$, so by Lemma 7 and the above notation, (6) becomes

$$\widetilde{M}(x)=\sum_{q\ge1}q\varphi(q)F\left(\frac{x}{q}\right)c(q)^2,\qquad c(q):=\sum_{\substack{k\ge1\\ q\mid k^2}}\frac{\mu(k)}{k^2}. \tag{8}$$

Since $\mu(k)\neq0$ only when k is squarefree, $c(q)=0$ unless q is cube-free. Hence it has the unique decomposition

$$q=ab^2,\qquad a,b \text{ squarefree},\qquad (a,b)=1.$$

For such q,

$$c(ab^2)=\sum_{\substack{k\ge1\\ ab\mid k}}\frac{\mu(k)}{k^2}=\frac{\mu(ab)}{a^2b^2}\sum_{\substack{m\ge1\\(m,ab)=1}}\frac{\mu(m)}{m^2}=\frac{\mu(ab)}{a^2b^2\zeta(2)}\prod_{p\mid ab}\left(1-\frac{1}{p^2}\right)^{-1}.$$

Define the multiplicative function B on positive integers by

$$B(n):=\prod_{p\mid n}\frac{p^3}{(p-1)(p+1)^2}.$$

Then using the identity $\varphi(q)=q\prod_{p\mid q}(1-1/p)$, we get

$$q\varphi(q)c(q)^2=\frac{\mu^2(ab)}{a^2\zeta(2)^2}B(ab),$$

and hence

$$M(x)=\frac{1}{\zeta(2)^2}\sum_{a\ge1}\frac{\mu^2(a)B(a)}{a^2}\sum_{\substack{b\ge1\\(a,b)=1}}\mu^2(b)B(b)F\left(\frac{x}{ab^2}\right). \tag{9}$$

For fixed squarefree a, write

$$\sum_{\substack{b\ge1\\(a,b)=1}}\frac{\mu^2(b)B(b)}{b^s}=\zeta(s)H_a(s),\qquad H_a(s)=\sum_{r\ge1}\frac{h_a(r)}{r^s}.$$

Then

$$H_a(s)=\prod_{p\mid a}\left(1-\frac{1}{p^s}\right)\prod_{p\nmid a}\left(1+\frac{B(p)}{p^s}\right)\left(1-\frac{1}{p^s}\right),$$

and therefore

$$h_a(p^j)=\begin{cases}-1, & p\mid a,\ j=1,\\ 0, & p\mid a,\ j\ge2,\\ B(p)-1, & p\nmid a,\ j=1,\\ -B(p), & p\nmid a,\ j=2,\\ 0, & p\nmid a,\ j\ge3.\end{cases} \tag{10}$$

In particular,

$$\sum_{r\ge 1}\frac{|h_a(r)|}{r}\ll\prod_{p\mid a}\left(1+\frac{1}{p}\right). \tag{11}$$

We shall also use the following bound.

Lemma 8. *Uniformly for squarefree $a\ge 1$ and $X\ge 1$,*

$$\sum_{r\le X}|h_a(r)|\ll\tau(a)X^{\frac12}.$$

Proof. Since h_a is multiplicative, we may separate the prime factors of r according as they divide a or not. Write

$$r=r_1r_2,\qquad (r_1,a)=1,\qquad p\mid r_2\Longrightarrow p\mid a.$$

By (10), if $p\mid a$ then $h_a(p)=-1$ and $h_a(p^j)=0$ for $j\ge 2$. Hence any r_2 in the support of h_a is squarefree and divides a. In particular, there are at most $\tau(a)$ possible choices for r_2, and $|h_a(r_2)|=1$. We now consider r_1. For $p\nmid a$, (10) shows that the exponent of p in an integer in the support of h_a is at most 2. Thus every such r_1 can be written uniquely as $r_1=uv^2$, where u and v are squarefree and $(u,v)=1$; the primes dividing u occur to the first power and those dividing v to the second power. Moreover,

$$|h_a(r_1)|=\prod_{p\mid u}|B(p)-1|\prod_{p\mid v}B(p).$$

The formula for $B(p)$ gives $0<B(p)<1$ and $0<1-B(p)\le 1/p$, so

$$|h_a(r_1)|\le\frac{\mu^2(u)}{u}\mu^2(v).$$

Therefore

$$\sum_{\substack{r_1\le X\\(r_1,a)=1}}|h_a(r_1)|\ll\sum_{uv^2\le X}\frac{\mu^2(u)}{u}\mu^2(v)\le\sum_{u\le X}\frac{\mu^2(u)}{u}\sum_{v\le (X/u)^{1/2}}1\ll X^{\frac12}\sum_{u\ge 1}\frac{\mu^2(u)}{u^{3/2}}\ll X^{\frac12}.$$

Finally, summing over the at most $\tau(a)$ possible values of r_2 gives the desired conclusion. □

Since

$$\sum_{\substack{b\ge 1\\(a,b)=1}}\frac{\mu^2(b)B(b)}{b^s}=\zeta(s)H_a(s),\qquad H_a(s)=\sum_{r\ge 1}\frac{h_a(r)}{r^s},$$

comparison of Dirichlet coefficients gives

$$\mu^2(b)B(b)\mathbb{1}_{(a,b)=1}=\sum_{r\mid b}h_a(r). \tag{12}$$

Indeed, the right-hand side is the Dirichlet convolution $\mathbf{1}*h_a$, whose Dirichlet series is $\zeta(s)H_a(s)$.

We now use (12) to rewrite $\widetilde{M}(x)$ in a form adapted to the square variable b^2.

Lemma 9. *Let*

$$f(t):=\sum_{a\ge 1}\sum_{r\ge 1}\frac{\mu^2(a)B(a)h_a(r)}{a^2}F\left(\frac{1}{atr^2}\right),\qquad t>0. \tag{13}$$

Then f is absolutely convergent and

$$\widetilde{M}(x)=\frac{1}{\zeta(2)^2}\sum_{e\ge 1}f\left(\frac{e^2}{x}\right). \tag{14}$$

Proof. Substituting (12) into (9), we obtain

$$\begin{aligned}\widetilde{M}(x)&=\frac{1}{\zeta(2)^2}\sum_{a\ge 1}\frac{\mu^2(a)B(a)}{a^2}\sum_{b\ge 1}\left(\sum_{r\mid b}h_a(r)\right)F\left(\frac{x}{ab^2}\right)\\&=\frac{1}{\zeta(2)^2}\sum_{a\ge 1}\frac{\mu^2(a)B(a)}{a^2}\sum_{r,e\ge 1}h_a(r)F\left(\frac{x}{ar^2e^2}\right),\end{aligned}$$

where in the second line we introduced the variable $b = re$. Since

$$F\left(\frac{x}{ar^2e^2}\right) = F\left(\frac{1}{a(e^2/x)r^2}\right),$$

the last expression is precisely the right-hand side of (14). We justify the rearrangement by establishing absolute convergence. Since $F(u) \ll \min(1, u)$, we have, uniformly for $Y > 0$,

$$\sum_{e\geq 1} \min\left(1, \frac{Y}{e^2}\right) \ll Y^{\frac{1}{2}}.$$

Therefore

$$\sum_{a,r,e\geq 1} \frac{\mu^2(a)B(a)|h_a(r)|}{a^2} F\left(\frac{x}{ar^2e^2}\right) \ll x^{\frac{1}{2}} \sum_{a\geq 1} \frac{\mu^2(a)B(a)}{a^{5/2}} \sum_{r\geq 1} \frac{|h_a(r)|}{r}.$$

By (11),

$$\sum_{r\geq 1} \frac{|h_a(r)|}{r} \ll \prod_{p|a} \left(1 + \frac{1}{p}\right) \leq \tau(a),$$

and hence the last expression is $\ll x^{\frac{1}{2}} \sum_{a\geq 1} \tau(a)/a^{5/2} < \infty$. For each $t > 0$, the series defining $f(t)$ also converges absolutely, since $F(u) \leq u$ gives

$$\sum_{a,r\geq 1} \frac{\mu^2(a)B(a)|h_a(r)|}{a^2} F\left(\frac{1}{atr^2}\right) \leq \frac{1}{t} \sum_{a\geq 1} \frac{\tau(a)}{a^3} < \infty.$$

This completes the proof. □

We next record the estimates on f that will allow us to evaluate the e-sum in (14).

Lemma 10. *The function f defined in* (13) *is continuous on $(0, \infty)$ and satisfies*

$$f(t) \ll \begin{cases} t^{-1/4}, & 0 < t \leq 1, \\ t^{-1}, & t \geq 1. \end{cases} \tag{15}$$

The same bounds hold true for the sum in (13) *with absolute value on each term.*

Proof. We first consider $t \geq 1$. Since $F(u) \ll u$,

$$|f(t)| \ll \frac{1}{t} \sum_{a\geq 1} \frac{\mu^2(a)B(a)}{a^3} \sum_{r\geq 1} \frac{|h_a(r)|}{r^2} \ll \frac{1}{t} \sum_{a\geq 1} \frac{\tau(a)}{a^3} \ll t^{-1},$$

where we used (11). Suppose now that $0 < t \leq 1$. We first restrict to $a \leq t^{-1}$ and put $D_a := (at)^{-1/2} \geq 1$. For $r \leq D_a$, we use $F \leq 1$ and Lemma 8 to obtain

$$\sum_{a\leq t^{-1}} \frac{1}{a^2} \sum_{r\leq D_a} |h_a(r)| \ll \sum_{a\leq t^{-1}} \frac{\tau(a) D_a^{1/2}}{a^2} = t^{-1/4} \sum_{a\leq t^{-1}} \frac{\tau(a)}{a^{9/4}} \ll t^{-1/4}.$$

For $r > D_a$, we use

$$F\left(\frac{1}{atr^2}\right) \ll \frac{1}{atr^2}.$$

By partial summation and Lemma 8,

$$\sum_{r>D_a} \frac{|h_a(r)|}{r^2} \ll \tau(a) D_a^{-3/2}.$$

Consequently, this range contributes

$$\ll \frac{1}{t} \sum_{a\leq t^{-1}} \frac{1}{a^3} \tau(a) D_a^{-3/2} = t^{-1/4} \sum_{a\leq t^{-1}} \frac{\tau(a)}{a^{9/4}} \ll t^{-1/4}.$$

Finally, if $a > t^{-1}$, then $at > 1$, so

$$F\left(\frac{1}{atr^2}\right) \ll \frac{1}{atr^2}.$$

Thus this range contributes

$$\ll \frac{1}{t} \sum_{a > t^{-1}} \frac{\tau(a)}{a^3} \ll 1 \ll t^{-1/4}.$$

This proves (15). The same estimates, uniformly for t in compact subsets of $(0, \infty)$, show that the series defining f converges locally uniformly. Since each summand is continuous in t, the function f is continuous. □

We can now pass from the sum over squares in (14) to an integral.

Lemma 11. *For $x \geq 1$, we have*

$$\begin{aligned} \sum_{e \geq 1} f\left(\frac{e^2}{x}\right) &= \sqrt{x} \int_0^\infty f(t)\, d\left(t^{1/2}\right) + O\left(x^{1/3}\right) \\ &= \frac{\sqrt{x}}{2} \int_0^\infty \frac{f(t)}{t^{1/2}}\, dt + O\left(x^{1/3}\right). \end{aligned} \tag{16}$$

Proof. For $Y > 0$, put

$$S(Y) := \sum_{e \geq 1} F\left(\frac{Y}{e^2}\right), \qquad C_F := \int_0^\infty F\left(u^{-2}\right)\, du = \frac{1}{2} \int_0^\infty \frac{F(y)}{y^{3/2}}\, dy.$$

The integral converges since $F(y) \ll \min(1, y)$. We first show that, uniformly for $Y > 0$,

$$S(Y) = C_F \sqrt{Y} + O\left(Y^{1/3}\right). \tag{17}$$

Suppose $Y \geq 1$, and let $N \geq 1$ be an integer. Set $G_Y(u) := F(Y/u^2)$. Since F is bounded, the portions of the sum and integral below N contribute $O(N)$ to their discrepancy. Since $F(t) = \{t\}(1 - \{t\})$ is globally Lipschitz and satisfies $|F'(t)| \leq 1$ for $t \notin \mathbb{Z}$, the chain rule gives

$$|G_Y'(u)| = \frac{2Y}{u^3} \left|F'\left(\frac{Y}{u^2}\right)\right| \leq \frac{2Y}{u^3}$$

for almost every $u > 0$. Comparing each summand $G_Y(e)$ with its integral over $[e-1, e]$, and using the fundamental theorem of calculus, gives

$$\left| \sum_{e > N} G_Y(e) - \int_N^\infty G_Y(u)\, du \right| \leq \int_N^\infty |G_Y'(u)|\, du \leq \frac{Y}{N^2}.$$

Since $\int_0^\infty G_Y(u)\, du = C_F \sqrt{Y}$, we obtain

$$S(Y) = C_F \sqrt{Y} + O\left(N + \frac{Y}{N^2}\right).$$

Taking $N = \lceil Y^{1/3} \rceil$ proves (17) for $Y \geq 1$. If $0 < Y < 1$, then $S(Y) \ll Y$, and the same estimate follows from $Y + \sqrt{Y} \ll Y^{1/3}$. We now apply this estimate term by term. The absolute convergence established in Lemma 9 gives

$$\sum_{e \geq 1} f\left(\frac{e^2}{x}\right) = \sum_{a, r \geq 1} \frac{\mu^2(a) B(a) h_a(r)}{a^2} S\left(\frac{x}{ar^2}\right).$$

To sum the errors, write $A_a(D) := \sum_{r \leq D} |h_a(r)|$. By partial summation and Lemma 8,

$$\begin{aligned} \sum_{r \geq 1} \frac{|h_a(r)|}{r^{2/3}} &= \frac{2}{3} \int_1^\infty \frac{A_a(D)}{D^{5/3}}\, dD \\ &\ll \tau(a) \int_1^\infty D^{-7/6}\, dD \ll \tau(a). \end{aligned} \tag{18}$$

Consequently, the total error in (17) is bounded by

$$x^{1/3}\sum_{a\geq 1}\frac{\mu^2(a)B(a)}{a^{7/3}}\sum_{r\geq 1}\frac{|h_a(r)|}{r^{2/3}} \ll x^{1/3}\sum_{a\geq 1}\frac{\tau(a)}{a^{7/3}} \ll x^{1/3}.$$

It follows that

$$\sum_{e\geq 1} f\left(\frac{e^2}{x}\right) = C_F\sqrt{x}\sum_{a,r\geq 1}\frac{\mu^2(a)B(a)h_a(r)}{a^{5/2}r} + O\left(x^{1/3}\right).$$

Finally, the change of variables $y = 1/(atr^2)$ gives

$$\frac{1}{2}\int_0^\infty F\left(\frac{1}{atr^2}\right)\frac{dt}{t^{1/2}} = \frac{C_F}{a^{1/2}r}.$$

We may sum this identity over a, r by (11). Thus

$$\frac{1}{2}\int_0^\infty \frac{f(t)}{t^{1/2}}\,dt = C_F\sum_{a,r\geq 1}\frac{\mu^2(a)B(a)h_a(r)}{a^{5/2}r},$$

which proves the lemma. □

Combining Lemma 9 and Lemma 11, we obtain

$$\widetilde{M}(x) = \frac{\sqrt{x}}{2\zeta(2)^2}\int_0^\infty \frac{f(t)}{t^{1/2}}\,dt + O_d(x^{1/3}). \tag{19}$$

We now substitute the definition of f given in (13) into (19); one may interchange summation and integration due to Lemma 10. Put

$$I := \int_0^\infty \frac{F(y)}{y^{3/2}}\,dy < \infty.$$

The change of variables $y = 1/atr^2$ gives

$$\int_0^\infty F\left(\frac{1}{atr^2}\right)\frac{dt}{t^{1/2}} = \frac{1}{a^{1/2}r}\int_0^\infty \frac{F(y)}{y^{3/2}}\,dy = \frac{I}{a^{1/2}r}. \tag{20}$$

Therefore

$$\begin{aligned}\widetilde{M}(x) &= \frac{\sqrt{x}}{2\zeta(2)^2}\sum_{a,r\geq 1}\frac{\mu^2(a)B(a)h_a(r)}{a^2}\int_0^\infty F\left(\frac{1}{atr^2}\right)\frac{dt}{t^{1/2}} + O_d(x^{1/3})\\ &= \frac{\sqrt{x}}{2\zeta(2)^2}\left(\int_0^\infty \frac{F(y)}{y^{3/2}}\,dy\right)\sum_{a\geq 1}\frac{\mu^2(a)B(a)}{a^{5/2}}\sum_{r\geq 1}\frac{h_a(r)}{r} + O_d(x^{1/3}).\end{aligned} \tag{21}$$

Indeed, the corresponding absolute sum is bounded by

$$I\sum_{a\geq 1}\frac{\mu^2(a)B(a)}{a^{5/2}}\sum_{r\geq 1}\frac{|h_a(r)|}{r} \ll \sum_{a\geq 1}\frac{\tau(a)}{a^{5/2}} < \infty.$$

It remains to evaluate the factors in (21); we begin with evaluation of the integral, which can be thought of as an Archimedean factor.

Lemma 12. *We have*

$$\int_0^\infty \frac{F(y)}{y^{3/2}}\,dy = \frac{2}{\pi}\zeta\left(\frac{3}{2}\right). \tag{22}$$

Proof. First, we use the Fourier expansion [Apo76, Theorem 12.19],

$$\{y\}(1-\{y\}) = \frac{1}{\pi^2}\sum_{n\geq 1}\frac{1-\cos(2\pi ny)}{n^2}.$$

This expansion has non-negative summands, hence Tonelli's theorem gives

$$\int_0^\infty \frac{F(y)}{y^{3/2}}\,dy = \frac{1}{\pi^2}\sum_{n\geq 1}\frac{1}{n^2}\int_0^\infty \frac{1-\cos(2\pi ny)}{y^{3/2}}\,dy.$$

By a classical integral identity [GR07, Eq. (3.823), $\mu = -1/2$, $a = 1/2$],

$$\int_0^\infty \frac{1-\cos u}{u^{3/2}}\,du = \sqrt{2\pi}$$

and scaling $u = 2\pi ny$, we obtain

$$\int_0^\infty \frac{1-\cos(2\pi ny)}{y^{3/2}}\,dy = 2\pi n^{1/2}.$$

Thus

$$\int_0^\infty \frac{F(y)}{y^{3/2}}\,dy = \frac{2}{\pi}\sum_{n\geq 1}\frac{1}{n^{3/2}} = \frac{2}{\pi}\zeta\left(\frac{3}{2}\right).$$

□

We turn to the a-sum in (21), which we express as an Euler product. By the definition of H_a,

$$\sum_{r\geq 1}\frac{h_a(r)}{r} = H_a(1) = \prod_{p\mid a}\left(1-\frac{1}{p}\right)\prod_{p\nmid a}\left(1+\frac{B(p)}{p}\right)\left(1-\frac{1}{p}\right).$$

Since $\mu^2(a)$ restricts a to squarefree integers, summing over a gives an Euler product. At a fixed prime p, there are two cases. If $p \nmid a$, the local factor is

$$\left(1-\frac{1}{p}\right)\left(1+\frac{B(p)}{p}\right),$$

whereas if $p \mid a$, the factor contributed by $\mu^2(a)B(a)a^{-5/2}H_a(1)$ is

$$\frac{B(p)}{p^{5/2}}\left(1-\frac{1}{p}\right).$$

Combining these two cases gives

$$\sum_{a\geq 1}\frac{\mu^2(a)B(a)}{a^{5/2}}H_a(1) = \prod_{p}\left(1-\frac{1}{p}\right)\left(1+\frac{B(p)}{p}+\frac{B(p)}{p^{5/2}}\right). \tag{23}$$

Substituting (22) and (23) into (21) proves the quantitative formula for $M(x)$ in Theorem 3, and therefore both variance asymptotics by Proposition 2.

Acknowledgments

This research is supported by the Israel Science Foundation (grant no. 2088/24) and an Alon Fellowship.

References

[Apo76] Tom M. Apostol. *Introduction to Analytic Number Theory.* Undergraduate Texts in Mathematics. Springer-Verlag, New York, 1976.

[GR07] I. S. Gradshteyn and I. M. Ryzhik. *Table of Integrals, Series, and Products.* Academic Press, 7 edition, 2007.

[Hal82] R. R. Hall. Squarefree numbers on short intervals. *Mathematika*, 29:7–17, 1982.

[Sim26] Omer Simhi. Squarefree values of random polynomials are gaussian mixtures, 2026. preprint.

[Sof26] Efthymios Sofos. Counting square-free values of random polynomials. *arXiv preprint arXiv:2601.19319*, 2026.